Use-Case Flow Specification

Disaster Relief Supply & Volunteer Coordinator | Problem Statement #64 | Lab 1 – PES University

Use Case	UC-02: Allocate Supplies & Set Delivery Priority
Primary Actor	Disaster Manager
Secondary Actor	Shelter Coordinator (notified of the allocation)
Includes / Extended by	Includes UC-03 Check Warehouse Stock. Extended by UC-04 Flag Supply Shortfall (when stock is short).
Related Requirements	FR-001, FR-002, NFR-002
Trigger	Disaster Manager opens the dispatch queue and chooses "Allocate Supplies".

Preconditions

1. Disaster Manager is registered, verified and logged in.
2. At least one requisition is Open and already has a priority (FR-001).
3. Central warehouse stock data is available (live or last synced).

Postconditions (on success)

1. Each processed requisition is marked Allocated (or Partially Allocated) and a manifest line exists for it.
2. Warehouse stock is reduced by exactly the quantity allocated.
3. The manifest is ordered by dispatch priority and ready for volunteer and vehicle dispatch (UC-05).
4. An audit-log entry is written and each Shelter Coordinator is notified.

Main Success Scenario

1. Disaster Manager logs in and opens the dispatch queue.
2. System lists Open requisitions ordered P1, P2, P3 with shelter, item, quantity and hours of stock left.
3. Manager selects "Allocate Supplies".
4. System runs Check Warehouse Stock (UC-03) for each item category in the queue.
5. System allocates in priority order: each requisition receives the smaller of the quantity requested and the stock remaining, and stock is deducted, all in one atomic transaction.
6. System creates manifest lines in dispatch order and sets each requisition to Allocated.
7. System writes the audit-log entry and notifies each Shelter Coordinator.
8. System displays the logistics manifest and the Manager can continue to dispatch volunteers (UC-05). Use case ends.

Alternate Flow A1 – Insufficient stock (at step 5)

1. System finds that remaining stock for an item is less than a requisition requests.
2. System gives the remaining stock to the higher-priority requisitions first and marks the others Partially Allocated or Not Allocated.
3. System invokes Flag Supply Shortfall (UC-04): it records the missing quantity per requisition and alerts the Manager.
4. Flow resumes at step 6 with partial or empty manifest lines. Use case ends.

Alternate Flow A2 – Stock changed by another allocation (at step 5)

1. Another manager's allocation run commits first, so this transaction cannot complete.
2. System rolls back, reloads the latest stock and requisitions, and shows "Stock changed – review and retry".
3. Manager returns to step 2.

Business Rules

P1 before P2 before P3; ties broken by people affected, then oldest request. Water or medical kits with 24 hours or less of stock are always P1. Stock is never allocated twice. Only a verified Disaster Manager may allocate. Every change is audit-logged.